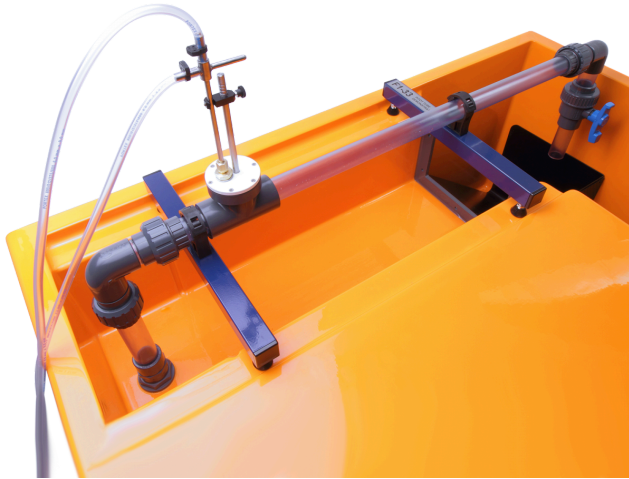




F1-33 Pitot Tube Demonstrator

The pitot tube can be moved across the cross-section of the pipe in order to measure the dynamic head profile.

PRODUCT CODE: F1-33

Categories: Educational, F1 Series Fluid Mechanics, F1-33 - Pitot Tube Demonstrator

Tags: Armfield , F1-33 , Fluid Mechanics , Fluid Velocity , Hydraulics Bench , Pitot Tube , Pitot Tube Demonstrator , Pressurised Water Manometer , Static Head , Static Tube , Total Head , Velocity Flow Profile

DESCRIPTION

The pitot tube can be moved across the cross-section of the pipe in order to measure the dynamic head profile. The position of the measuring tip relative to the wall of the pipe can be read on a scale.

The pitot tube is connected to a pressurised water manometer to measure the differential head across the pitot static tube.

TECHNICAL SPECIFICATIONS

Inside diameter of test pipe: 27mm

Pitot-static tube outside diameter: 6mm

Pitot-static tube inside diameter: 3.2mm

Scale length of manometer tubes: 500mm

Cross section of manometer tubes: 5.6mm diameter

Range of pitot-static tube traverse: 21mm with 3mm scale increments

Requires Hydraulics Bench Service unit F1-10/F1-10-2

FEATURES & BENEFITS

Experimental content

- Demonstrating the use of a pitot-static tube to measure the dynamic head produced by water flowing inside a pipe using a pressurised water manometer to measure the difference between the static head and the total head
- Demonstrating the relationship between static head, total head and dynamic head
- Demonstrating how a pitot-static tube can be used to determine the velocity of a fluid
- Demonstrating how the dynamic head of a fluid flowing inside a pipe varies with radius due to the development of a boundary layer at the wall of the pipe

- Demonstrating how the dynamic head profile varies at the entrance to a pipe downstream of a 90 degree bend with undeveloped flow

Dimensions

Length: 1.00m

Width: 0.35m

Height: 0.52m

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